
HTML Tutorial

Ref: <https://www.w3schools.com/> and
other internet based tutorials

Goal of this tutorial

- Not to teach everything about HTML, but provide the basic knowledge
 - Explain code of examples
 - Provide some useful references
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What is HTML?

- HTML stands for Hyper Text Markup Language
 - It describes the structure of a Web page
 - HTML consists of a series of elements
 - Elements tell the browser how to display the content
 - Elements are represented by tags
 - HTML tags label pieces of content such as "heading", "paragraph", "table", and so on
 - Browsers do not display the HTML tags, but use them to render the content of the page
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Example

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>

<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

HTML basic structure

- The `<!DOCTYPE html>` declaration defines this document to be HTML5.
- In older documents (HTML 4 or XHTML), the declaration is more complicated because the declaration must refer to a DTD (Document Type Definition).
 - HTML 4.01:
 - `<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">`
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the document
- The `<title>` element specifies a title for the document
- The `<body>` element contains the visible page content
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph

HTML Tags

- HTML tags are element names surrounded by angle brackets:
<tagname>content goes here...</tagname>
- HTML tags normally come in pairs like <p> and </p>
- The first tag in a pair is the start tag, the second tag is the end tag
- The end tag is written like the start tag, but with a forward slash inserted before the tag name
- Tip: The start tag is also called the opening tag, and the end tag the closing tag.

Web Browsers

- The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them.
- The browser does not display the HTML tags, but uses them to determine how to display the document:

HTML Basics

■ HTML Documents

- The HTML document itself begins with `<html>` and ends with `</html>`.
- The visible part of the HTML document is between `<body>` and `</body>`.

■ Example

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

HTML Basics

■ HTML Headings

- HTML headings are defined with the `<h1>` to `<h6>` tags.
- `<h1>` defines the most important heading. `<h6>` defines the least important heading:

■ Example

- `<h1>This is heading 1</h1>`
- `<h2>This is heading 2</h2>`
- `<h3>This is heading 3</h3>`

■ HTML Paragraphs

- HTML paragraphs are defined with the `<p>` tag:

■ Example

- `<p>This is a paragraph.</p>`

HTML Basics

■ HTML Links

- HTML links are defined with the `<a>` tag:

■ Example

- `<a href="https://www.w3schools.com">This is a link</a>`
- The link's destination is specified in the href attribute.
- Attributes are used to provide additional information about HTML elements.

■ HTML Images

- HTML images are defined with the `<img>` tag.
- The source file (src), alternative text (alt), width, and height are provided as attributes:

■ Example

- ``

HTML Basics

■ HTML Buttons

- HTML buttons are defined with the `<button>` tag:

■ Example

- `<button>Click me</button>`

■ HTML Lists

- HTML lists are defined with the `<ul>` (unordered/bullet list) or the `<ol>` (ordered/numbered list) tag, followed by `<li>` tags (list items):

■ Example

`<ul>`

`<li>Coffee</li>`

`<li>Tea</li>`

`</ul>`

`<ol>`

`<li>Coffee</li>`

`<li>Tea</li>`

`</ol>`

HTML <head>

- The HTML <head> element is a container for metadata. HTML metadata is data about the HTML document. Metadata is not displayed.
- The <head> element is placed between the <html> tag and the <body> tag:
- Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First HTML</title>
  </head>
  <body>
```

HTML Elements and Attributes

■ HTML Elements

- An HTML element usually consists of a start tag and an end tag, with the content inserted in between:
 - `<tagname>Content goes here...</tagname>`
- The HTML element is everything from the start tag to the end tag:
 - `<p>My first paragraph.</p>`
- HTML elements with no content are called empty elements. Empty elements do not have an end tag, such as the `<br>` element (which indicates a line break).
- Some HTML elements will display correctly, even if you forget the end tag:
- Example

```
<html>
<body>
<p>This is a paragraph
</body>
</html>
```

HTML Elements and Attributes (cont...)

■ HTML Attributes

- All HTML elements can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**

■ Example

- HTML links are defined with the <a> tag. The link address is specified in the href attribute:

```
<a href="https://www.w3schools.com">This is a link</a>
```

- The filename of the image source is specified in the src attribute:

```

```

- The style attribute is used to specify the styling of an element, like color, font, size etc.

```
<p style="color:red">This is a paragraph.</p>
```

HTML Elements and Attributes (cont...)

- All HTML elements can have attributes
 - The title attribute provides additional "tool-tip" information
 - The href attribute provides address information for links
 - The width and height attributes provide size information for images
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Example of Some Attributes

Attribute	Description
alt	Specifies an alternative text for an image, when the image cannot be displayed
href	Specifies the URL (web address) for a link
id	Specifies a unique id for an element
src	Specifies the URL (web address) for an image
style	Specifies an inline CSS style for an element
title	Specifies extra information about an element (displayed as a tool tip)

HTML Paragraphs

- HTML Paragraphs

- The HTML `<p>` element defines a paragraph:

- The Poem Problem

- This poem will display on a single line:

- Example

- `<p>`

My Bonnie lies over the ocean.

My Bonnie lies over the sea.

My Bonnie lies over the ocean.

Oh, bring back my Bonnie to me.

`</p>`

- The HTML <pre> Element

- The HTML <pre> element defines preformatted text.
- The text inside a <pre> element is displayed in a fixed-width font (usually Courier), and it preserves both spaces and line breaks:

- Example

- <pre>
- My Bonnie lies over the ocean.
- My Bonnie lies over the sea.
- My Bonnie lies over the ocean.
- Oh, bring back my Bonnie to me.
- </pre>

HTML Styles

- Use the *style* attribute for styling HTML elements
- Use *background-color* for background color
- Use *color* for text colors
- Use *font-family* for text fonts
- Use *font-size* for text sizes
- Use *text-align* for text alignment
- Example
 - `<h1 style="color:blue;">This is a heading</h1>`
 - `<h1 style="text-align:center;">Centered Heading</h1>`
 - `<h1 style="font-size:300%;">This is a heading</h1>`

HTML Formatting Elements

- HTML also defines special elements for defining text with a special meaning.
- HTML uses elements like `<b>` and `<i>` for formatting output, like bold or italic text.
- Formatting elements were designed to display special types of text:
- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Small text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

HTML Colors

- HTML colors are specified using predefined color names, or RGB, HEX, HSL, RGBA, HSLA values.

- Background Color

- `<p style="background-color:Tomato;">Lorem ipsum...</p>`

- Text Color

- `<h1 style="color:Tomato;">Hello World</h1>`

- The color "Tomato" can also be represented as :

- ❑ `rgb(255, 99, 71)`

- ❑ `#ff6347`

- ❑ `hsl(9, 100%, 64%)`

- Setting of border color

- `<h1 style="border:2px solid Tomato;">Hello World</h1>`

- Same as color name "Tomato", but 50% transparent:

- ❑ `rgba(255, 99, 71, 0.5)`

- ❑ `hsla(9, 100%, 64%, 0.5)`

■ HSL Value

- In HTML, a color can be specified using hue, saturation, and lightness (HSL) in the form:
 - `hsl(hue, saturation, lightness)`
- Hue is a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue.
- Saturation is a percentage value, 0% means a shade of gray, and 100% is the full color.
- Lightness is also a percentage, 0% is black, 50% is neither light or dark, 100% is white

HTML Block and Inline Elements

- Every HTML element has a default display value depending on what type of element it is.
- The two display values are: block and inline.

- Block-level Elements

- A block-level element always starts on a new line and takes up the full width available (stretches out to the left and right as far as it can).
- The `<div>` element is a block-level element.
- `<div style="background-color:black;color:white;padding:20px;">`
- `<h2>London</h2>`
- `<p>London is the capital city of England. </p> </div>`

- Inline Elements

- An inline element does not start on a new line and only takes up as much width as necessary.
- This is an inline `<span>` element inside a paragraph.

- Example

- `<h1>My <span style="color:red">Important</span> Heading</h1>`s